R13

Code No: 115AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
PRINCIPLES OF PROGRAMMING LANGUAGES
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) What are the factors that influence the basic design of programming languages?
b) What are the advantages and disadvantages of implementing a language with pure interpreter and with a compiler? [8+7]

2.a) In what way do operational semantics differ from denotational semantics?
b) Write a denotational semantics mapping function for "for" statement in Ada and Java. [7+8]

3.a) Discuss the design and evaluation of enumeration and subrange types with relevant examples.
b) Discuss about guarded commands. [8+7]

4.a) Discuss various types of assignment statements.
b) Explain the situation where multiple selection using if construct is more efficient than switch-case construct. [7+8]

5.a) What are the distinct semantic models for formal parameters. Explain.
b) Explain about co-routines with examples. [7+8]

6.a) Describe the deep-access method of implementing dynamic scoping.
b) Give a detailed note on pass-by-value and pass-by-result parameter passing methods. [8+7]

7.a) What is meant by subprogram level concurrency? Explain.
b) Write a brief note on Java threads. [8+7]

8. Explain about storage control and types of binding supported by Python. [15]

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